

# CACIC 2026 Supplementary Material

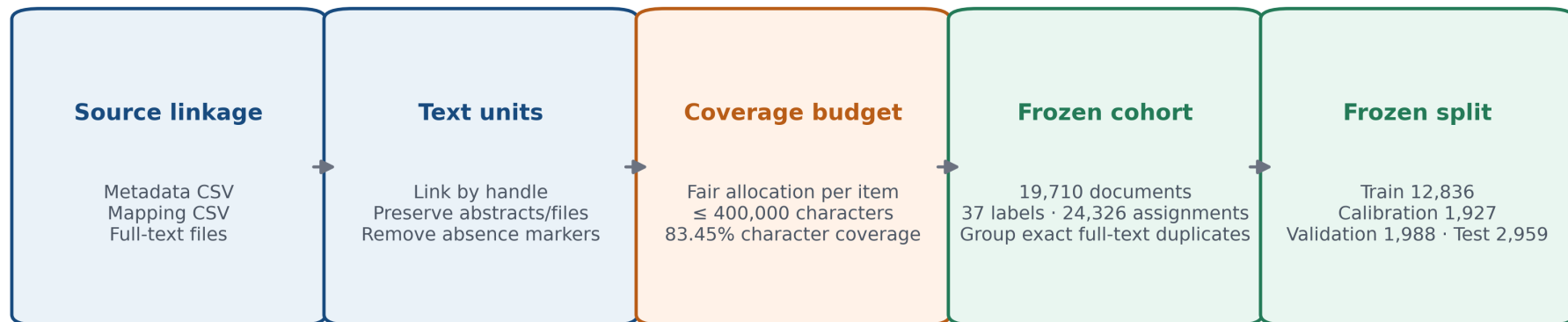
## Beyond Embeddings: Language-Aware Sparse Representations and Fine-Tuned Multilingual Transformers

Aggregate reviewer evidence only. Repository texts, handles, item-level predictions, split manifests, model weights, credentials, and personal paths are excluded.

### S1. Corpus and protocol

| Property                                   | Value                          |
|--------------------------------------------|--------------------------------|
| Documents                                  | 19710                          |
| Labels / assignments                       | 37 / 24326                     |
| Abstract segments / full-text files        | 24411 / 20750                  |
| Items with multiple abstracts / full texts | 4365 / 463                     |
| Represented full-text characters           | 83.45%                         |
| Train / calibration / validation / test    | 12,836 / 1,927 / 1,988 / 2,959 |

### Dataset construction and leakage-controlled partitioning



All 37 labels occur in every partition; representations are fitted on train only.

Figure S1. Dataset construction and frozen partitions.

## Calibrated multilingual long-document evaluation protocol

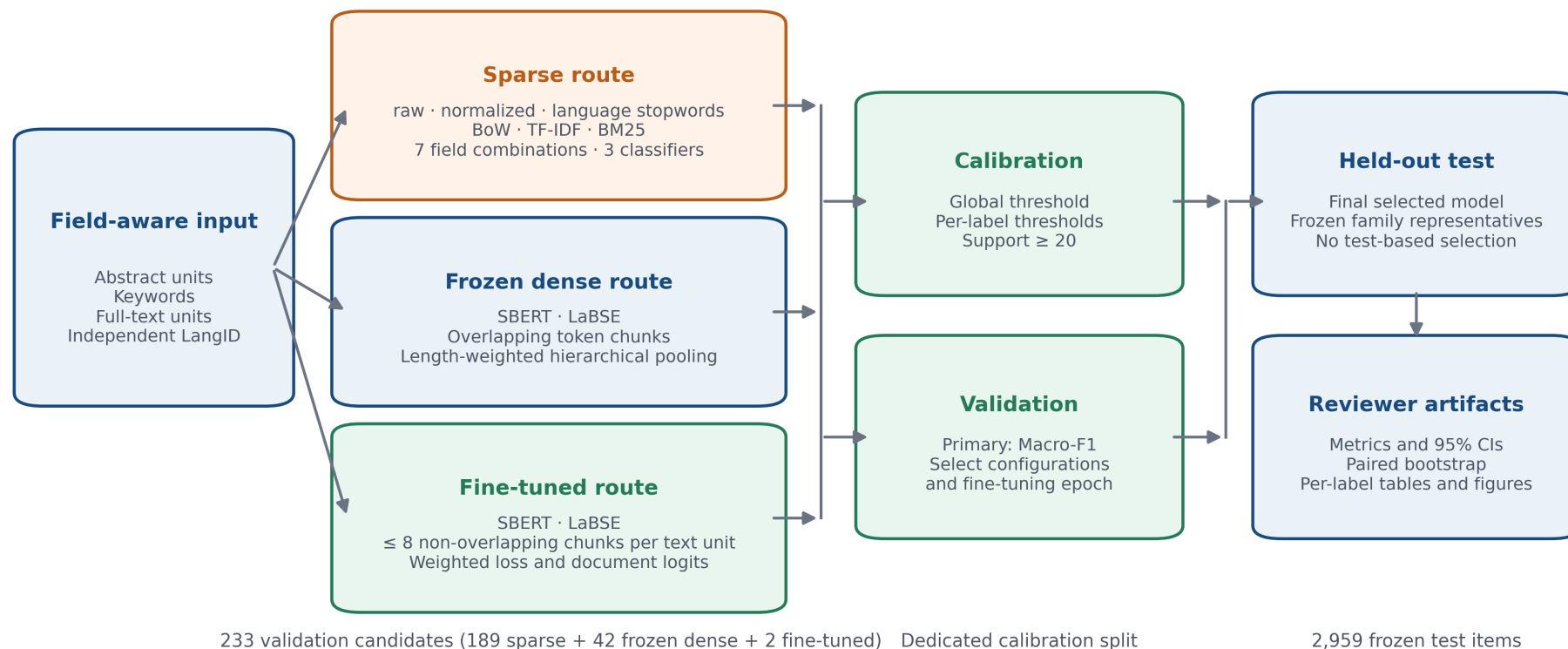


Figure S2. Experimental and selection protocol.

## S2. Complete experiment overview

| Rank | Configuration                                                   | Macro-F1 | Micro-F1 |
|------|-----------------------------------------------------------------|----------|----------|
| 1    | bm25 / fulltext / language_stopwords / linear_svc               | 0.7483   | 0.7705   |
| 2    | bm25 / abstract+keywords+fulltext / language_stopwords / logreg | 0.7388   | 0.7774   |
| 3    | tfidf / abstract+fulltext / language_stopwords / linear_svc     | 0.7357   | 0.7571   |
| 4    | bm25 / fulltext / raw / linear_svc                              | 0.7338   | 0.7480   |
| 5    | bm25 / fulltext / normalized / linear_svc                       | 0.7335   | 0.7484   |
| 6    | tfidf / fulltext / language_stopwords / linear_svc              | 0.7329   | 0.7550   |
| 7    | tfidf / keywords+fulltext / normalized / linear_svc             | 0.7316   | 0.7533   |

| Rank | Configuration                                                        | Macro-F1 | Micro-F1 |
|------|----------------------------------------------------------------------|----------|----------|
| 8    | tfidf / abstract+keywords+fulltext / language_stopwords / linear_svc | 0.7315   | 0.7559   |
| 9    | bm25 / abstract+fulltext / language_stopwords / logreg               | 0.7312   | 0.7752   |
| 10   | tfidf / abstract+keywords+fulltext / raw / linear_svc                | 0.7296   | 0.7509   |
| 11   | bm25 / keywords+fulltext / language_stopwords / logreg               | 0.7295   | 0.7704   |
| 12   | tfidf / abstract+keywords+fulltext / normalized / linear_svc         | 0.7294   | 0.7512   |
| 13   | tfidf / keywords+fulltext / raw / linear_svc                         | 0.7294   | 0.7517   |
| 14   | bm25 / abstract+keywords+fulltext / language_stopwords / linear_svc  | 0.7291   | 0.7612   |
| 15   | tfidf / keywords+fulltext / language_stopwords / linear_svc          | 0.7291   | 0.7543   |
| 16   | bm25 / fulltext / language_stopwords / logreg                        | 0.7283   | 0.7718   |
| 17   | bm25 / abstract+keywords+fulltext / language_stopwords / sgd         | 0.7276   | 0.7706   |
| 18   | tfidf / fulltext / raw / linear_svc                                  | 0.7254   | 0.7490   |
| 19   | bm25 / keywords+fulltext / language_stopwords / sgd                  | 0.7251   | 0.7648   |
| 20   | tfidf / fulltext / normalized / linear_svc                           | 0.7249   | 0.7482   |

Complete 233-row validation table: [https://github.com/KharolusIII/institutional-repository-subject-classification/blob/main/paper\\_artifacts/runs/v3\\_main/results\\_validation\\_grid.csv](https://github.com/KharolusIII/institutional-repository-subject-classification/blob/main/paper_artifacts/runs/v3_main/results_validation_grid.csv)

## S3. Family-level held-out comparison

| Family          | Selected feature set       | Validation Macro-F1 | Test Macro-F1 | Test Micro-F1 |
|-----------------|----------------------------|---------------------|---------------|---------------|
| bm25            | fulltext                   | 0.7483              | 0.7457        | 0.7641        |
| tfidf           | abstract+fulltext          | 0.7357              | 0.7338        | 0.7529        |
| labse_finetuned | abstract+keywords+fulltext | 0.7148              | 0.7069        | 0.7279        |
| bow             | abstract+keywords+fulltext | 0.6895              | 0.6782        | 0.7135        |
| sbert_finetuned | abstract+keywords+fulltext | 0.6663              | 0.6595        | 0.6746        |
| sbert_frozen    | keywords+fulltext          | 0.6278              | 0.6211        | 0.6397        |
| labse_frozen    | keywords+fulltext          | 0.6254              | 0.6143        | 0.6334        |

## S4. Convergence audit

| Configuration                                                | Macro-F1 | Micro-F1 |
|--------------------------------------------------------------|----------|----------|
| bm25 / fulltext / language_stopwords / linear_svc_c05_tol5e4 | 0.7497   | 0.7732   |
| bm25 / fulltext / language_stopwords / linear_svc_c05_tol1e4 | 0.7497   | 0.7732   |
| bm25 / fulltext / language_stopwords / linear_svc_c1_tol5e4  | 0.7483   | 0.7705   |

| Configuration                                                 | Macro-F1 | Micro-F1 |
|---------------------------------------------------------------|----------|----------|
| bm25 / fulltext / language_stopwords / linear_svc_c1_tol1e4   | 0.7483   | 0.7705   |
| tfidf / fulltext / language_stopwords / linear_svc_c1_tol1e4  | 0.7329   | 0.7550   |
| tfidf / fulltext / language_stopwords / linear_svc_c1_tol5e4  | 0.7329   | 0.7550   |
| tfidf / fulltext / language_stopwords / linear_svc_c05_tol1e4 | 0.7307   | 0.7554   |
| tfidf / fulltext / language_stopwords / linear_svc_c05_tol5e4 | 0.7307   | 0.7554   |

## S5. Final held-out result

| Metric                 | Estimate        | 95% CI           |
|------------------------|-----------------|------------------|
| Macro-F1               | 0.7522          | [0.7372, 0.7643] |
| Micro-F1               | 0.7685          | [0.7567, 0.7797] |
| Subset accuracy        | 0.5863          | —                |
| Hamming loss           | 0.0164          | —                |
| Precision@3 / Recall@3 | 0.3810 / 0.9320 | —                |

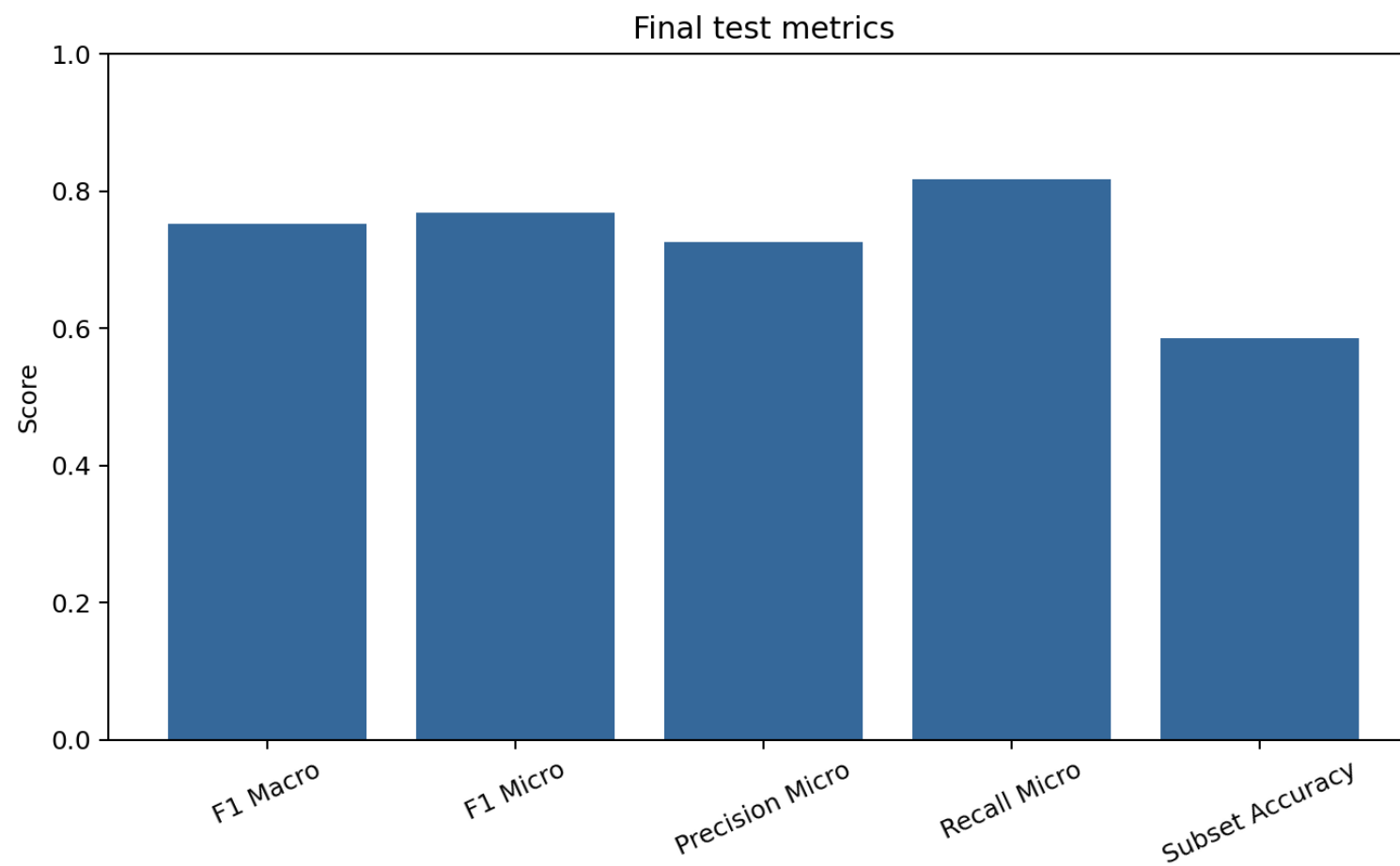


Figure S3. Final test metrics.

## S6. Fine-tuned transformer validation

| Model           | Feature set                | Selected epoch | Macro-F1 | Micro-F1 |
|-----------------|----------------------------|----------------|----------|----------|
| labse_finetuned | abstract+keywords+fulltext | 5              | 0.7148   | 0.7321   |
| sbert_finetuned | abstract+keywords+fulltext | 5              | 0.6663   | 0.6794   |

## S7. Paired bootstrap comparisons

| Candidate       | Metric   | Candidate – final BM25 | 95% CI             | P(candidate > BM25) |
|-----------------|----------|------------------------|--------------------|---------------------|
| tfidf           | f1_macro | -0.0184                | [-0.0299, -0.0068] | 0.0000              |
| tfidf           | f1_micro | -0.0156                | [-0.0243, -0.0072] | 0.0010              |
| labse_finetuned | f1_macro | -0.0453                | [-0.0587, -0.0311] | 0.0000              |
| labse_finetuned | f1_micro | -0.0406                | [-0.0508, -0.0303] | 0.0000              |
| bow             | f1_macro | -0.0741                | [-0.0896, -0.0590] | 0.0000              |
| bow             | f1_micro | -0.0549                | [-0.0662, -0.0441] | 0.0000              |
| sbert_finetuned | f1_macro | -0.0928                | [-0.1070, -0.0779] | 0.0000              |
| sbert_finetuned | f1_micro | -0.0939                | [-0.1055, -0.0822] | 0.0000              |
| sbert_frozen    | f1_macro | -0.1311                | [-0.1458, -0.1163] | 0.0000              |
| sbert_frozen    | f1_micro | -0.1288                | [-0.1398, -0.1175] | 0.0000              |
| labse_frozen    | f1_macro | -0.1379                | [-0.1528, -0.1233] | 0.0000              |
| labse_frozen    | f1_micro | -0.1350                | [-0.1469, -0.1226] | 0.0000              |

## S8. Per-subject results

| Subject                    | Support | Precision | Recall | F1     | Average precision |
|----------------------------|---------|-----------|--------|--------|-------------------|
| Odontología                | 66      | 0.9275    | 0.9697 | 0.9481 | 0.9792            |
| Ciencias Informáticas      | 387     | 0.9623    | 0.9225 | 0.9420 | 0.9746            |
| Educación Física           | 73      | 0.9211    | 0.9589 | 0.9396 | 0.9882            |
| Relaciones Internacionales | 115     | 0.8433    | 0.9826 | 0.9076 | 0.9674            |
| Psicología                 | 100     | 0.8774    | 0.9300 | 0.9029 | 0.9597            |
| Ciencias Veterinarias      | 72      | 0.7753    | 0.9583 | 0.8571 | 0.9307            |
| Letras                     | 182     | 0.7907    | 0.9341 | 0.8564 | 0.9442            |

| Subject                  | Support | Precision | Recall | F1     | Average precision |
|--------------------------|---------|-----------|--------|--------|-------------------|
| Bibliotecología          | 37      | 0.8611    | 0.8378 | 0.8493 | 0.9084            |
| Ciencias Agrarias        | 116     | 0.7754    | 0.9224 | 0.8425 | 0.8997            |
| Bellas Artes             | 106     | 0.7388    | 0.9340 | 0.8250 | 0.8927            |
| Ciencias Jurídicas       | 154     | 0.7245    | 0.9221 | 0.8114 | 0.9085            |
| Ciencias Económicas      | 85      | 0.9672    | 0.6941 | 0.8082 | 0.9138            |
| Ciencias Médicas         | 54      | 0.7385    | 0.8889 | 0.8067 | 0.8280            |
| Sociología               | 165     | 0.7622    | 0.8545 | 0.8057 | 0.8616            |
| Ingeniería               | 88      | 0.7207    | 0.9091 | 0.8040 | 0.8744            |
| Arquitectura             | 77      | 0.8852    | 0.7013 | 0.7826 | 0.8961            |
| Química                  | 62      | 0.7105    | 0.8710 | 0.7826 | 0.8316            |
| Filosofía                | 53      | 0.9714    | 0.6415 | 0.7727 | 0.8262            |
| Geografía                | 35      | 1.0000    | 0.6286 | 0.7719 | 0.8621            |
| Ciencias Naturales       | 186     | 0.7022    | 0.8495 | 0.7689 | 0.8314            |
| Física                   | 62      | 0.7143    | 0.8065 | 0.7576 | 0.8395            |
| Ciencias Astronómicas    | 37      | 0.6471    | 0.8919 | 0.7500 | 0.8160            |
| Historia                 | 124     | 0.6522    | 0.8468 | 0.7368 | 0.8067            |
| Antropología             | 59      | 0.6970    | 0.7797 | 0.7360 | 0.8191            |
| Astronomía               | 35      | 1.0000    | 0.5714 | 0.7273 | 0.9160            |
| Trabajo Social           | 46      | 0.7619    | 0.6957 | 0.7273 | 0.8109            |
| Ciencias de la Educación | 50      | 0.7447    | 0.7000 | 0.7216 | 0.7747            |
| Zoología                 | 33      | 0.6486    | 0.7273 | 0.6857 | 0.6669            |
| Periodismo               | 88      | 0.6311    | 0.7386 | 0.6806 | 0.7561            |
| Humanidades              | 227     | 0.5545    | 0.8062 | 0.6571 | 0.7860            |
| Comunicación Social      | 93      | 0.6275    | 0.6882 | 0.6564 | 0.7186            |
| Comunicación             | 103     | 0.5769    | 0.7282 | 0.6438 | 0.6717            |
| Educación                | 185     | 0.5437    | 0.7405 | 0.6270 | 0.6475            |
| Ciencias Exactas         | 126     | 0.7765    | 0.5238 | 0.6256 | 0.7233            |
| Urbanismo                | 30      | 0.4340    | 0.7667 | 0.5542 | 0.6012            |
| Ciencias Sociales        | 104     | 0.5316    | 0.4038 | 0.4590 | 0.4887            |
| Biología                 | 34      | 0.2152    | 0.5000 | 0.3009 | 0.2084            |

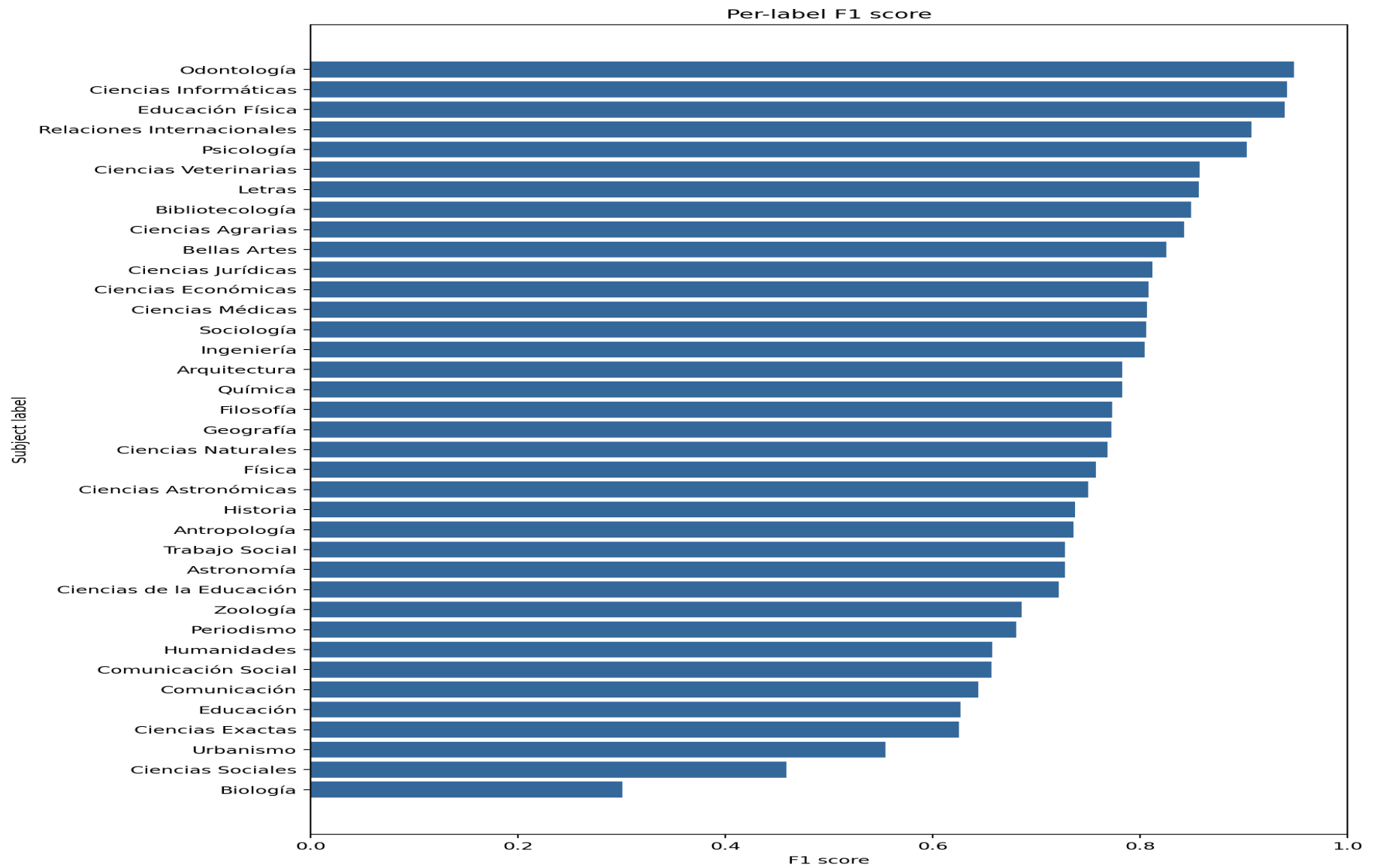


Figure S4. F1 by subject.



Figure S5. One-vs-rest confusion matrices.

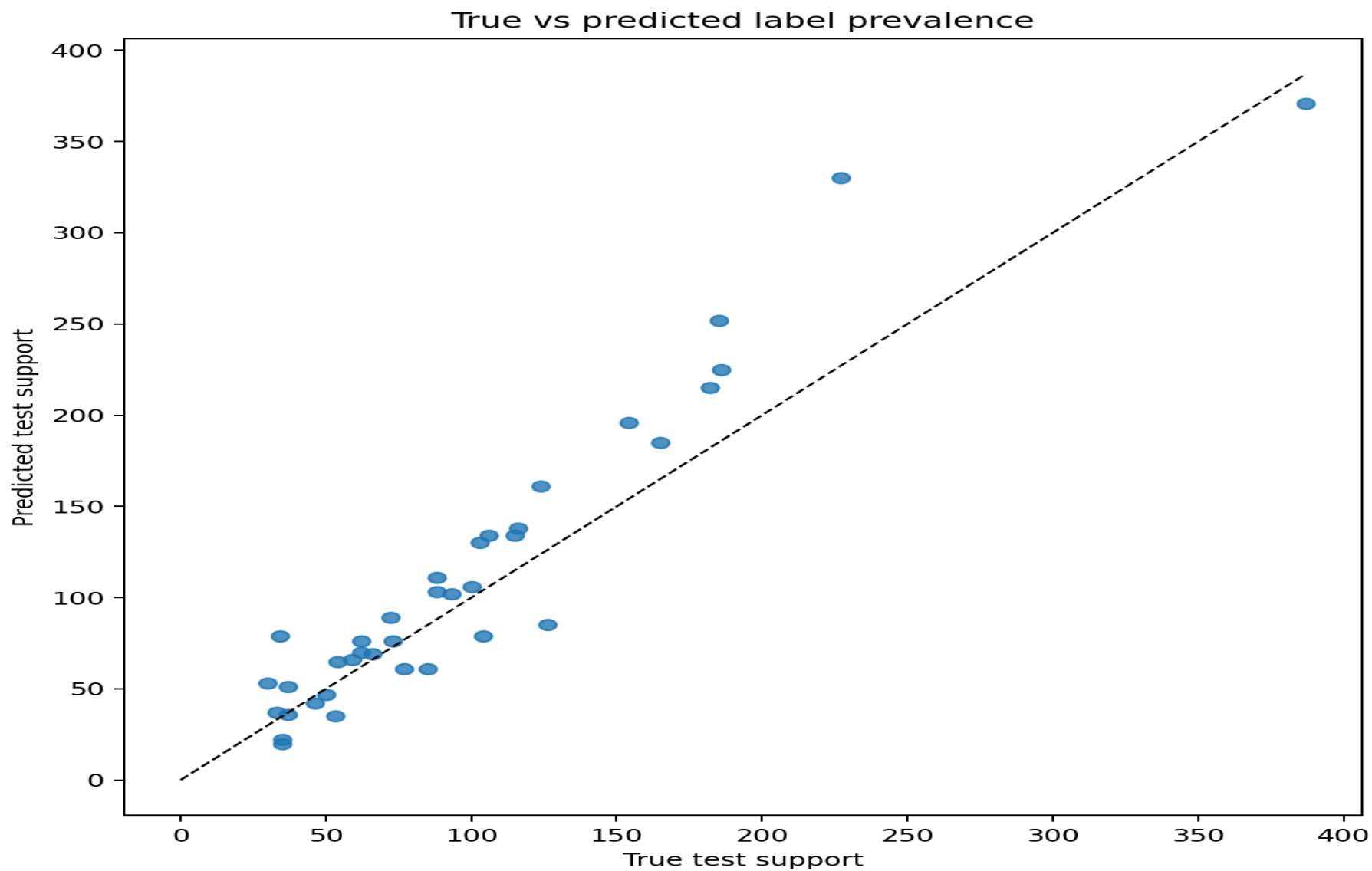


Figure S6. True versus predicted support.

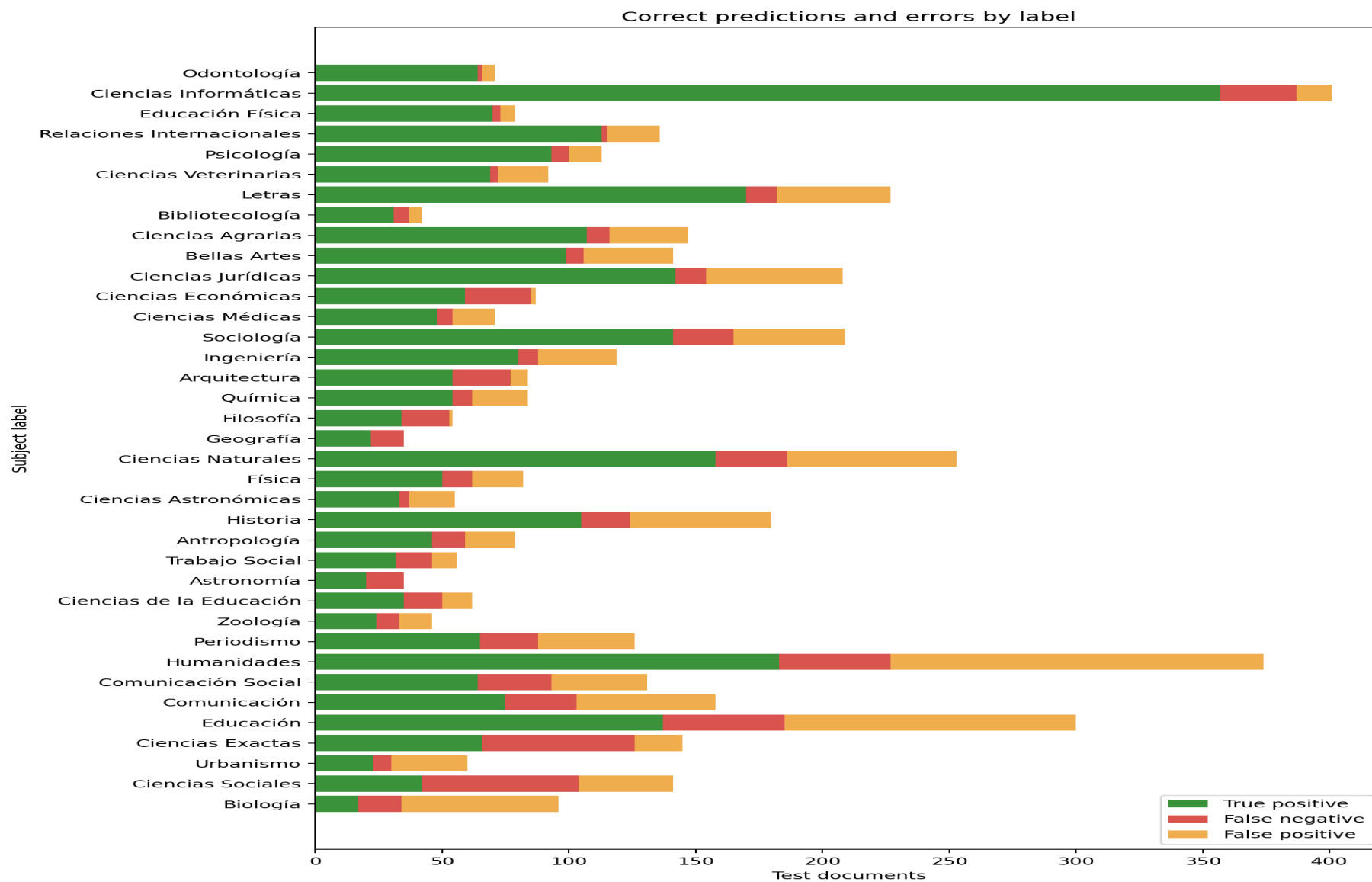


Figure S7. False-positive and false-negative counts.

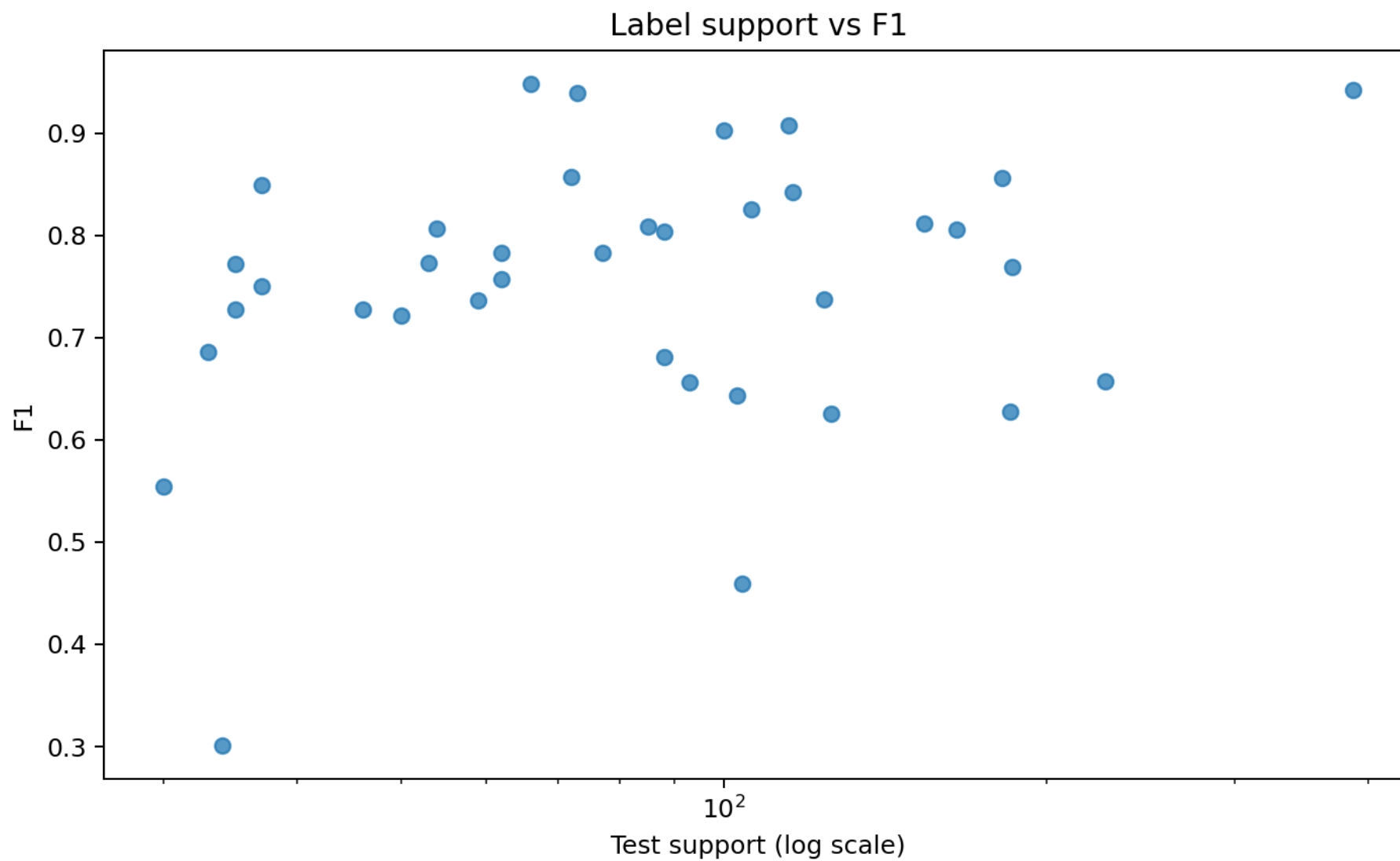


Figure S8. Test support versus subject-level F1.

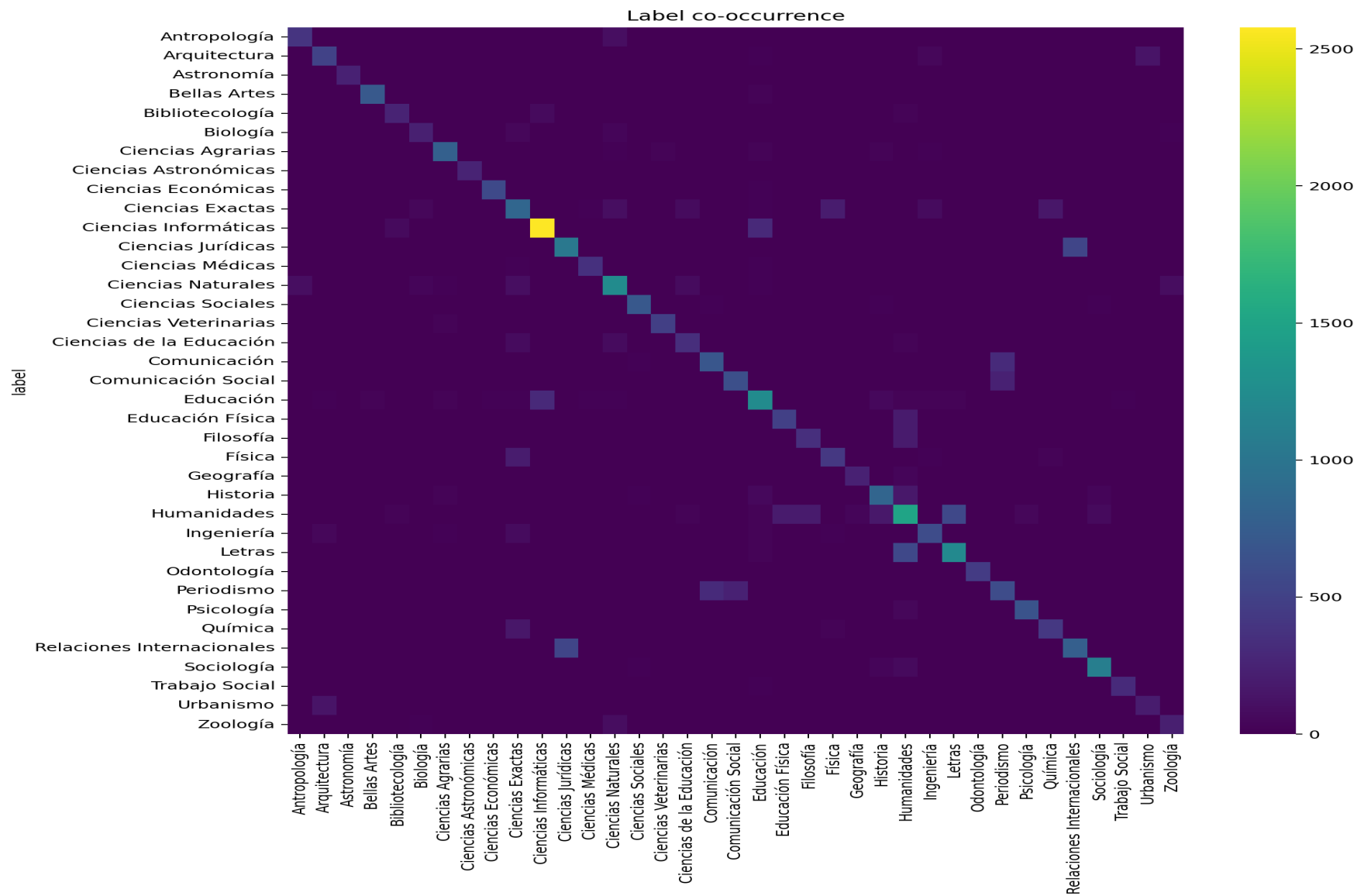


Figure S9. Label co-occurrence structure.

## S9. Reproducibility package

Complete aggregate CSV evidence, figures, source configurations, the executable public notebook, and SHA-256 manifest are available at [https://github.com/KharolusIII/institutional-repository-subject-classification/tree/main/paper\\_artifacts](https://github.com/KharolusIII/institutional-repository-subject-classification/tree/main/paper_artifacts)